

MANGALORE UNIVERSITY



National Education Policy – 2020 [NEP-2020]

Question Bank FOR

VI SEMESTER BCA

FUNDAMENTALS OF DATA SCIENCE WEB CONTENT MANAGEMENT SYSTEM

Fundamentals of Data Science

Question Bank

Unit 1

2 marks questions

- 1) Define data mining.
- 2) What is knowledge discovery in databases?
- 3) List any four stages of KDD.
- 4) What is Loosely Coupled DBMS and Tightly Coupled DBMS?
- 5) What is prediction and description in Data Mining?
- 6) List any four discovery driven tasks.
- 7) Define Support and Confidence.
- 8) Write the objectives of clustering.
- 9) Define Rough Set.
- 10) What is sequence mining and spatial data mining?
- 11) Write the subtasks of web mining.
- 12) Expand KDT and IE.
- 13) What is web mining and text mining?
- 14) List any two issues and challenges in data mining.
- 15) List the two application areas of data mining.
- 16) Explain the use of Data Visualization?
- 17) Define Machine Learning?
- 18) Define Supervised and Unsupervised Learning?
- 19) Explain the two fundamental goals of Data Mining?
- 20) Define Discovery Model? Give an example?
- 21) What is Frequent Episodes? List its types?
- 22) What is Deviation Detection? Give example?

Long Answer questions

- 1) Write the difference between KDD and Data Mining. (4)
- 2) Define KDD process. Explain the different stages of KDD. (6)
- 3) Distinguish between DBMS and Data Mining. (4)
- 4) Explain interaction of Data Mining Systems with Relational DBMS (4)
- 5) Explain the following Data Mining techniques. (4/6)
i) Verification Model ii) Discovery Model
- 6) Write a note on i) Discovery of Association Rules ii) Discovery of Classification Rules (4/6)
- 7) Write a note on i) Clustering ii) Frequent Episodes (4/6)
- 8) Write a note on i) Genetic Algorithms ii) Rough Set Techniques (4/6)
- 9) Write a note on Support Vector Machines (SVM) (4)
- 10) Explain any two discovery driven tasks. (4/6)
- 11) Write a note on the following: (6)
i) Sequence Mining ii) Web Mining iii) Spatial Data Mining
- 12) List and explain the issues and challenges in Data Mining. (4/6)
- 13) Explain the different application areas of Data Mining. (6)

Unit 2

2 marks questions

- 1) What is data warehouse?
- 2) How are organizations using the information from data warehouses?
- 3) Expand OLTP and OLAP.
- 4) What is data cube? How it is defined?
- 5) Define dimensions and facts.
- 6) Define base cuboid and apex cuboid.
- 7) Define measure and mention any two categories.
- 8) Define concept hierarchy. Give an example.
- 9) Write any two differences between OLAP system and Statistical database.
- 10) What is Starnet Model? Give an example through diagram.
- 11) List any two benefits of Business analyst by having data warehouse.
- 12) List any two data warehouse models from the architecture point of view.
- 13) List any four back-end tools and utilities included in data warehouse.
- 14) Define data cleaning and data integration.
- 15) Define data transformation and data reduction.
- 16) What is Dimensionality Reduction? Mention any one method used for Dimensionality Reduction.
- 17) List any two methods for data discretization.
- 18) Write the difference between Operational database and Data Warehouse?
- 19) Define ROLAP and MOLAP?
- 20) What is Data mart? List its types?
- 21) Explain Virtual Warehouse?
- 22) What type of information a Metadata Repository should include?
- 23) What is Metadata Repository?
- 24) Explain the approaches to data cleaning as a process?
- 25) Define Binning?
- 26) What are the use of Data scrubbing tool and Data Auditing tool?
- 27) What is Entity Identification Problem?
- 28) List the processes involved in data transformation?
- 29) What is the use of Attribute Subset Selection?
- 30) What is Histograms? Mention the use of Multidimensional histograms.

Long answer questions

- 1) Define and explain data warehouse. (4/6)
- 2) Explain any four/six differences between operational database system and data warehouse. (4/6)
- 3) Explain the following Multidimensional database schema with example. (3 marks each)
 - i.) Star schema
 - ii.) Snowflake schema
 - iii.) Fact constellation schema
- 4) Define Measure. Explain different categories of measures. (4)
- 5) Explain any four/six OLAP operations. (4/6)
- 6) Explain different views regarding the business analysis framework design of a data warehouse. (4)
- 7) Explain the steps in data warehouse design process. (4)
- 8) Explain the three-tier architecture of Data warehouse with neat diagram. (6)
- 9) Explain the different data warehouse models from the architecture point of view. (4)

- 10) Explain the different back-end tools and utilities included in data warehouse. (4)
- 11) What is metadata? Explain metadata repository. (6)
- 12) Explain different types of OLAP servers. (4/6)
- 13) Explain the different forms of data preprocessing. (4)
- 14) Explain (any four) the different ways of handling missing values. (4/6)
- 15) Define noise and explain different data smoothing techniques. (6)
- 16) Explain the different steps involved in data transformation. (4)
- 17) What is data reduction? Explain the strategies for data reduction. (6)
- 18) Write a note on Wavelet Transform and Principal Component Analysis. (6)
- 19) Explain any two numerosity reduction techniques. (4)
- 20) Explain any three methods for data discretization. (6)
- 21) Write a note on different methods used for the generation of concept hierarchies for categorical data. (4/6)
- 22) Explain Data cube aggregation? (4)
- 23) Explain heuristic methods of attribute subset selection with an example and explain its techniques. (6)
- 24) What is sampling? Explain the ways used to sample for data reduction?

UNIT III

2 marks questions:

- 1) What are frequent patterns? What is the use of frequent pattern mining?
- 2) What is market basket analysis?
- 3) What is the meaning of the following association rule:
Computer $\Rightarrow$ Antivirus_Software (Support=2%, Confidence=60%)
 - 4) What is relative support and absolute support?
 - 5) What is frequent itemset?
 - 6) What is closed itemset?
 - 7) Give the two step process of association rule mining.
 - 8) What is the purpose of sequential pattern mining and structured pattern mining.
 - 9) What is the purpose of Apriori property? Mention it.
 - 10) Mention the two step process followed in Apriori algorithm.
 - 11) List any 2 techniques used to improve the efficiency of Apriori based mining.
 - 12) What is the strategy of frequent pattern growth?
 - 13) What is horizontal data format and vertical data format?
 - 14) Expand ECLAT and ARCS.
 - 15) Name any two pruning strategies in mining closed frequent itemset.
 - 16) Write the difference between multilevel association rule and multidimensional association rule?
 - 17) What are categorical attributes and quantitative attributes?
 - 18) Name any two common binning strategies with respect to mining quantitative association rule?
 - 19) Name any two constraints included in constraint-based mining.
 - 20) How are metarules useful?
 - 21) Define frequent set and border set.
 - 22) What is FP growth?
 - 23) Define Support count or Frequency.
 - 24) Define Maximal Frequent itemset

Long answer questions:

- 1) Explain Market Basket Analysis with an example. (4/6)
- 2) Write a note on frequent itemsets and closed itemsets. (4)
- 3) Explain the classification of frequent pattern mining. (6)
- 4) Explain the two-step process followed in Apriori algorithm. (4)
- 5) Explain Apriori algorithm. (6)
- 6) Write a note on generating association rules from frequent itemsets. (4)
- 7) Explain any two variations to improve the efficiency of Apriori based mining. (4)
- 8) Explain any three variations to improve the efficiency of Apriori based mining. (6)
- 9) With diagram, explain mining by partitioning data. (4)
- 10) Explain any two pruning strategies of mining closed frequent itemsets. (4)
- 11) Explain the three pruning strategies of mining closed frequent itemsets. (6)
- 12) Write a note on: i) Mining Multilevel association rule with uniform minimum support
ii) Mining Multilevel association with reduced minimum support. (6)
- 13) Briefly explain the steps involved in ARCS. (6)
- 14) What is constrain based mining? Explain the constraints included in constrain based mining. (6)
- 15) Explain association rules with example.
- 16) Explain Apriori algorithm for finding frequent item sets with an example.
- 17) Explain support, confidence and lift measure with respect to association rule mining.
- 18) Explain the basic concept of frequent itemset mining.
- 19) Explain multilevel association rules and multidimensional association rules for mining data.

TID	T100	T200	T300	T400	T500	T600	T700	T800	T900
ITEM IDS	I1,I2,I5	I2,I4	I2,I3	I1,I2,I4	I1,I3	I2,I3	I1,I3	I1,I2,I3,I5	I1,I2,I3

Generate the list of frequent item-set ordered by their corresponding suffixes, where the minimum support count is 2 and minimum confidence is 60%.

- 20) Make use of the database which has five transactions. Let minimum support = 60% and minimum confidence = 80%.

Transaction	Items
T10	M, O, N, K, E, Y
T20	D, O, N, K, E, Y
T30	M, A, K, E
T40	M, U, C, K, Y
T50	C, O, O, K, I, E

Find all frequent item sets using Apriori and FP-growth, respectively

- 21) Explain about Constraint based Association mining
- 22) Describe the steps involved in improving the efficiency of the Apriori algorithm.

Unit IV

2 marks questions

- 1) What is Classification?
- 2) What is prediction?
- 3) What is regression Analysis?
- 4) Write steps involved in data classification?
- 5) What is supervised learning?
- 6) What is unsupervised learning?
- 7) What is correlation Analysis?
- 8) What is Decision tree induction?
- 9) What is Decision tree?
- 10) What is tree pruning?
- 11) What is the use of tree selection measures?
- 12) Expand CART, ID3
- 13) What is splitting criterion? List distinct splitting criteria's values.
- 14) What is attribute selection measure? List popular attribute measures.
- 15) What is information Gain? Write the formula to find expected information needed to classify a tuple.
- 16) What is entropy?
- 17) Define gain ratio. Write formula.
- 18) Define Gini index. Write formula.
- 19) List the use of MDL.
- 20) Write two approaches to tree pruning.
- 21) What is pessimistic pruning?
- 22) Expand and briefly explain BOAT.
- 23) What is eager learners? Give examples.
- 24) What is Instance based Learners?
- 25) What is clustering? Write the advantages.
- 26) What is the purpose of Portioning Algorithm?
- 27) Expand CLARA and ROCK.
- 28) Expand BIRCH and DBSCAN.
- 29) Expand OPTICS and DENCLUE.
- 30) What is STING clustering.
- 31) What is grid based clustering? Give examples.

II. Long Questions (4 to 6 marks)

1. Explain Decision tree induction algorithm.
2. Explain the criteria which are used for comparing classification and prediction methods.
3. List and explain the issues in classification and prediction methods.
4. Explain the Data Classification with example.
5. Explain the classification by Decision tree induction with an example.
6. Explain distinct values of splitting attribute based on training data in decision tree induction algorithm.
7. Explain Bayes' Theorem.
8. Explain Naïve Bayesian Classification.
9. Write a note on Bayesian Belief Network.
10. Write the steps involved in training Bayesian Belief Network.
11. Explain IF-THEN rules for classification.
12. Explain rule extraction from Decision tree.

13. Explain Sequential covering algorithm.
14. Explain K-nearest neighbours classification.
15. Explain linear regression.
16. Explain nonlinear regression.
17. Explain the requirements of clustering in data mining.
18. Explain K-means Partition Algorithm.
19. Explain K-mediod Partition Algorithm.
20. Explain 4 cases of cost function for k-mediod clustering.
21. Write a note on CLARA
22. Compare Agglomerative and Divisive Hierarchical Clustering.
23. Write a note on BIRCH
24. Write a note on ROCK.
25. Write a note on Chameleon.
26. Explain DBSCAN.
27. Explain OPTICS.
28. Explain DENCLUE.
29. Explain STING.
30. Explain Wave Cluster.

WEB CONTENT MANAGEMENT SYSTEM-

QUESTION BANK

UNIT - I

TWO MARKS QUESTIONS

1. What is content management system?
2. What are the common functions of CMS?
3. What are the types of content management system?
4. Define web content management System?
5. What are the types of WCMS?
6. Define a) CMA b) CDA
7. Define DAM.
8. What is the difference between WCMS and CMS.
9. What is content and content format?
10. Mention any four Text file formats.
11. Mention any four common graphic file formats.
12. Write various standards defined by Moving Picture Experts Group.
13. What is digital graphics?
14. What is raster Graphics?
15. What is vector graphics?
16. List advantages and disadvantages of raster graphics.
17. List advantages and disadvantages of vector graphics.
18. Why do you need to connect content types and formats?
19. List best practices for digital graphics
20. What is audio editing? Mention the use of audio editing.
21. Mention the types of audio editing.
22. What is cutting and fading?
23. What is fading and mixing?
24. What are the benefits of audio editing?
25. List and briefly explain any two audio editing tool.

LONG ANSWER QUESTIONS

1. Explain the various types of content management systems. (6)
2. Explain the features of web content management systems. (6)
3. What are the Advantages of using WCMS? (4)
4. What are the disadvantages of using WCMS? (4)
5. Explain the various popular content types. (4/6)
6. Explain the various types of content formats in WCMS. (4/6)
7. Explain various text file formats.(4/6)
8. Explain various graphics file formats.(4/6)
9. Explain various audio file formats.(4/6)
10. Write a note on content tools. (4/6)
11. How to create digital graphics? (6)
12. Explain needs and general guidelines for content development. (4)
13. Explain the content development guidelines according to Mick Wood. (6)
14. Explain steps involved in the audio editing process.(4)

UNIT – II

TWO MARKS QUESTIONS

1. What is Web Hoisting?
2. Mention the types of web hosting services.
3. What are the benefits of web hosting?
4. What is dedicated hosting? List advantages.
5. What is co-located hosting? List disadvantages.
6. List the factors affecting the cost of webhosting.
7. Mention any four company names offering web hosting services.
8. Define multimedia content.
9. Mention the basic elements of multimedia.
10. What are the benefits of multimedia content?
11. What is Prezi?
12. What is Haiku Deck?
13. What is Apple keynote?
14. What is wiki?
15. Write any two characteristics of Wiki.
16. Write any two examples of Wikis.
17. Expand WPML. list the steps to activate it.
18. Define multilingual content development.
19. What is Screen Casting?
20. Write any two use cases for screen casting.
21. Write features of Snagit.
22. Mention the four features of Microsoft PowerPoint.
23. Write examples for presentation software.
24. Write any four popular screen cast tools
25. What is multilingual content development?
26. Mention any two features of OBS studio.
27. What is the use of hreflang tag in HTML?
28. What are the different types of Email accounts?

LONG ANSWER QUESTIONS

1. List and explain the different features of web hosting? (4/6)
2. Why is web hosting important and explain any five features of web hosting services? (6)
3. Explain the different steps to host a website. (8)
4. Explain about how to upload website to your account. (4)
5. Mention and explain the best practices for creating multimedia content? (4/6)
6. Explain the basic elements of multimedia. (4)
7. Explain the different examples of presentation software that are commonly used in conjunction with WCMS platforms. (6)
8. Explain in detail the different steps to create a presentation using Google Slides. (6)
9. What are the different steps are involved in creating a wiki. (6)
10. How wikis work? (4/6)
11. What are the advantages of using wikis? (4/6)
12. Explain about advanced features and best practices in presentation software. (6)
13. Explain the key features of any three screen casting tools (6)
14. Write a note on Loom. (4/6)
15. Write any six key aspects of multilingual content development. (6)
16. Write the advantages of multilingual content development. (6)
17. Write and explain steps involved in developing multilingual content.
18. Explain the different steps are involved in creating multilingual content in Word Press.(6)

UNIT III

TWO MARKS QUESTIONS

1. Define dynamic website. Write example.
2. What is content generation?
3. Define static website. Write example.
4. Mention any two characteristics of dynamic website.

5. Mention any two characteristics of static website.
6. Differentiate static website and dynamic website.
7. Give two examples each for static and dynamic websites.
8. Expand AJAX. What is the use of it?
9. What is Wordpress?
10. What is Twitter?
11. Give four examples for server-side language.
12. Mention any two front-end development language and framework.
13. What is CSS? List types of CSS.
14. What are the 3 types of CSS?
15. What is the use of Type selector and Attribute selectors?
16. What is the use of class selector and child selectors?
17. Define a blog? Give examples.
18. Write any two purposes of blogs.
19. What is content sharing and Search Engine Optimization (SEO)?
20. How community sharing is facilitated through blogs?
21. What is monetization?
22. Write the steps to customize blogs.
23. Give two examples each of blog sites for i) Technology and Business and ii) Personal development and lifestyle.
24. Give two examples each of blog sites for – i) Parenting & Family and ii) Travel.

LONG ANSWER QUESTIONS

1. Explain the characteristics of dynamic websites. (4/6)
2. Explain the characteristics of static websites. (4/6)
3. Explain any four/six differences between static and dynamic websites. (4/6)
4. Write a note on any THREE dynamic websites. (6)
5. Illustrate the steps for creating dynamic web content. (6)
6. Explain selectors in CSS. (4)
7. Give detailed steps to create website using CSS. (6)
8. Write a note on the purposes for which blogs are used. (4/6)
9. Enumerate the steps to create your own blogs. (6)

UNIT – IV

TWO MARKS QUESTIONS

1. What is e-publishing? Give examples for e-publishing tools.
2. What is simulation? Write its purpose.
3. Write Characteristics of Simulation.
4. What are different types of simulation?
5. Write Application of Simulation.
6. What are the advantages of Simulation?
7. Define virtual reality. Write any two key feature.
8. Write application of virtual reality.
9. Write Two VR software categories with example.
10. Explain Babylon.js and FMOD studio.
11. What is the role of virtual reality in healthcare?
12. Write any two characteristics of 2D Animation.
13. What is Toon Boom Harmony? Write features.
14. What is Pencil2D? Write features.
15. What is Synfig Studio? Write features.
16. Write any TWO key characteristics of 3D animation.
17. How to perform texturing and rigging in 3D animation.
18. Write applications of 3D animation.
19. What is Autodesk Maya? Write features.
20. Differentiate 2D and 3D animation.
21. What is Moodle?

22. Define discussion forum.
23. Write features of discussion forum.
24. Mention any four discussion forums widely used online.
25. What is Joomla? List uses of Joomla.
26. Write Major versions of Joomla.
27. Mention any four websites created using Joomla.
28. What is Drupal? Write any two key features.
29. Differentiate Joomla and Drupal.
30. What is Quora?

LONG ANSWER QUESTIONS

1. Explain different models of e-publishing with its characteristics. (4/6)
2. Explain any six key categories of e-publishing. (6)
3. Explain various e-publishing tools in different categories. (6)
4. What are the steps to do e-publications? (6)
5. Explain different types of simulation. (4/6)
6. Explain characteristics of simulation. (4/6)
7. Write the Characteristics of Virtual Reality. (4)
8. Write a note on any two VR software categories. (6)
9. Write a note on Virtual Reality Experience. (6)
10. Differentiate between 2D and 3D animation. (4/6)
11. Explain any two popular 2D animation software tools. (4/6)
12. Write the steps to create animation using Pencil2D software. (6)
13. Write the steps to create 3D animations using Blender (6)
14. Write a note on any two 3D software applications (6)
15. Explain any six key features of Moodle. (6)
16. Explain the steps involved in creating a new course and uploading the content in Moodle. (6)
17. Explain the applications of Moodle. (5)
18. Write a short note on publishing course in Moodle. (6)
19. Explain the steps involved in Add a Quiz (Assessment) in Moodle (6)
20. Discuss the key features of discussion forums. (4)
21. Explain how Stack Overflow discussion forum works. (6)
22. Explain Why Joomla is used? (4/6)
23. Explain the steps involved to build website using Joomla. (6)
24. Explain features of Wordpress.
25. Explain the steps to create a website using WordPress. (6)
26. Explain the features of Drupal. (4/6)
27. Differentiate between WordPress and Joomla. (4/6)
28. Differentiate between Joomla and Drupal. (4/6)
29. Write the Steps to create website using Drupal. (6)
